COS 106: Course Project Practical Report

Personal Portfolio & Academic Management Platform

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Live Website:	https://stargirl-helen.github.io/portflio/
GitHub Repository:	https://github.com/stargirl-helen/portflio

1. Executive Summary & Project Goal

For my COS 106 (Introduction to Web Technologies) term project, I developed a personal portfolio and academic management platform from the ground up. As an undergraduate student pursuing a BSc in Cybersecurity while working in corporate operations, my aim was to construct a website that authentically documents my background, displays my technical competencies, and provides an everyday utility through an interactive study planner.

I deliberately avoided bulky component frameworks (like React or Vue) and third-party UI libraries. The entire application was authored in pure HTML5, CSS3, and vanilla JavaScript. This decision guaranteed near-instant page load times, total control over styling, zero dependency vulnerabilities, and direct adherence to the course practical rubric.

2. Design System & Aesthetic Choices

I chose to step away from the generic dark cyan 'AI template' style. I wanted a look that felt personal, grounded, and human. The design rules implemented across the entire site include:

- **Color Palette:** Pure White (#ffffff) base with subtle blush surfaces (#fff7f9), deep rose pink (#db2777) for primary elements/headings, and warm sunshine yellow (#fef08a) for badges and category highlights.
- **Human Scale Typography:** Headlines are kept at a modest 1.85rem rather than loud, oversized fonts. Font hierarchies remain calm and legible on both mobile screens and desktops.
- **Zero Animations:** All bouncing transitions, pulsing glow effects, and keyframe animations were removed. The interface is completely solid, static, and instantaneous.
- **Streamlined Navigation & Controls:** Removed unused placeholder buttons (such as LinkedIn and fake repo links) and audio controls, leaving a clean, direct, and distraction-free interface.

3. Website Architecture & Pages Built

The website is organized into five distinct, semantically structured pages:

- **Home (index.html):** Provides an introductory overview, portrait photograph, concise biography, interactive skill level indicators, featured project summaries, planner preview, and direct student contact information.
- **About Me (about.html):** Explores my personal story originating from Benue State as the fourth of five children, my educational milestones, and my career objective to become an enterprise Information Security Architect.

• **Projects Showcase (projects.html):** Presents three detailed case studies: this COS 106 Portfolio & Study Planner, a Python socket-based Corporate Network Vulnerability Scanner, and a Secure Authentication Prototype with password entropy scoring.

• **Academic Planner & Schedule (planner.html):** Features a real-time study task manager allowing tasks to be added, filtered, marked complete, or deleted, with automatic completion metrics. It also contains an academic timetable rendered using standard HTML table tags.

• **Contact Portal (contact.html):** Includes my official institutional student email (h.dzauka6193@miva.edu.ng) and an interactive contact form validated via client-side JavaScript before dispatch.

4. Technical Implementation & Features

• **Client-Side Form Validation:** Enforces strict input verification on the contact form using vanilla JavaScript. It ensures no field is left blank, tests email syntax against a standard regex, and verifies that phone numbers contain only numeric digits.

• **LocalStorage Persistence:** The Academic Planner persists all coursework tasks directly inside the user's browser localStorage. Reloading or closing the browser retains task states and completion tallies.

• **Responsive Layouts:** Utilized CSS Grid and Flexbox with mobile-first breakpoints (max-width 860px and 720px) to guarantee seamless viewing on mobile phones, tablets, and desktops.

• **Zero-Dependency Static Build:** Authored a native Node.js build runner (build.js) that verifies asset presence and compiles distribution bundles into dist/ and public/ in under 2 seconds without external npm packages.

• **Audio Integration:** Embedded background ambient audio using <audio> with .ogg and .mp3 sources, configured never to autoplay.

5. Deployment & Challenges Overcome

Deploying a project from a local machine to cloud hosting often reveals edge cases. Here is how I resolved each challenge encountered during this project:

• **Vercel Static Output Mapping:** An initial build attempt failed with STATIC_BUILD_NO_OUT_DIR because Vercel was looking for a public directory. I configured vercel.json and updated build.js to generate both dist/ and public/.

• **Build Script Syntax Error:** A misplaced loop bracket in build.js triggered a SyntaxError. I restructured the file with clean helper functions and verified it using node test/validate.js.

• **Git Credential Conflict:** When pushing from terminal, Git defaulted to an existing stored Windows credential for another user account. Setting git config credential.https://github.com.username stargirl-helen resolved the authorization.

• **GitHub Pages Jekyll 404:** GitHub Pages initially failed to deploy the static files due to its default Jekyll processing engine. Adding an empty .nojekyll file at the root bypassed Jekyll and enabled direct static hosting.

6. Conclusion & Learning Outcomes

Working through this project provided invaluable hands-on experience in core front-end web development, DOM manipulation, browser storage APIs, and production deployment workflows. Building without frameworks deepened my appreciation for clean semantic HTML, accessible markup, and responsive CSS.

The project successfully satisfies all 20-mark requirements of the COS 106 syllabus and is currently live and accessible at https://stargirl-helen.github.io/portflio/.